Mark schemes

Q1.

(a) (i) $(x - 2)^2$

Attempt to complete square for x

M1

centre has x-coordinate = 2

M1 implied if value correct or −2

A1

and y-coordinate = 0

Centre (2,0)

B1

3

(ii) RHS = 18

Withhold if circle equation RHS incorrect

B1

Radius = $\sqrt{18}$

Square root of RHS of equation (if > 0)

M1

Radius = $3\sqrt{2}$

A1

3

(b) Perpendicular bisects chord so need to use Length of 4

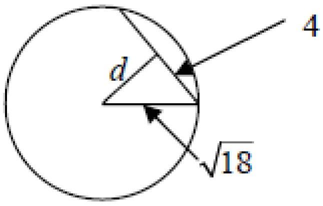
B1

d^2 = (radius)^2 − 4^2

M1

d^2 = 18 − 16 so perpendicular distance = $\sqrt{2}$

A1



3

(c) (i) $x^2 + (2k - x)^2 - 4x - 14 = 0$

M1

$(2k - x)^2 = 4k^2 - 4kx + x^2$

B1

$\Rightarrow 2x^2 + 4k^2 - 4kx - 4x - 14 = 0$

$(\Rightarrow x^2 + 2k^2 - 2kx - 2x - 7 = 0)$

$\Rightarrow x^2 - 2(k+1)x + 2k^2 - 7 = 0$

AG (be convinced about algebra and = 0)

A1

3

(ii) $4(k+1)^2 - 4(2k^2 - 7)$

"$b^2 - 4ac$" in terms of k (either term correct)

M1

$4k^2 - 8k - 32 = 0$ or $k^2 - 2k - 8 = 0$

$b^2 - 4ac = 0$ correct quadratic equation in k

A1

$(k - 4)(k + 2) = 0$

Attempt to factorise, solve equation

m1

$k = -2$, $k = 4$

SC B1, B1 for −2 , 4 (if M0 scored)

A1

4

(iii) Line is a tangent to the circle

Line touches circle at one point etc

E1

1

[17]

Q2.

(a) (i) Attempt at $\Delta y / \Delta x$ (used with numbers)

Not x over y

M1

$= \frac{2}{12} = \frac{1}{4}$

0.25 etc any correct equivalent

A1

2

(ii) $y - 2 = m(x - 11)$ or $y + 1 = m(x + 1)$

M1

$4y - x = -3$ etc leading to

$x - 4y = 3$

or y = mx+c and attempt to find c

(or sub both points into given equation)

AG (be convinced)

A1

2

(b) Attempt to eliminate x or y

17y = 17 etc

M1

$y = 1$

A1

$x = 7$

C is point (7,1)

A1

3

[7]

Q3.

(a) $(x - 6)^2 + (y - 3)^2$

B1

$= 36 + 9 - 20$

Generous with sign errors

M1

$= 5^2$

Condone 25

A1

3

(b) (i) Centre (6,3)

ft their a and b

B1✓

(ii) Radius = 5

Correct or ft $\sqrt{RHS}$ *if RHS > 0*

B1✓

2

(c) (i) $x^2 + (x + 4)^2 - 12x - 6(x + 4) + 20 = 0$

Or their $(x - a)^2 + (x + 4 - b)^2 = r^2$

M1

$(2x^2 - 10x + 12 = 0) \Rightarrow x^2 - 5x + 6 = 0$

AG (be convinced)

A1

2

(ii) $(x - 3)(x - 2) = 0$

Attempt at factors or use of formula

M1

$x = 2, \quad x = 3$

Both correct

A1

Substituting for one y value

m1

P, Q are (2,6) and (3,7)

Both points correct

A1

4

[11]

Q4.

(a) Midpoint $\left(\frac{6+2}{2}, \frac{5-1}{2}\right) = (4, 2)$

One coordinate correct unsimplified

M1

Both correct and simplified

A1

2

(b) $AB^2 = (6 - 2)^2 + (5 + 1)^2$

Pythagoras used (condone one slip)

M1

= (16 + 36) =52

52 or $\sqrt{52}$ seen

A1

$\Rightarrow AB = 2\sqrt{13}$

A1

3

(c) (i) Gradient $AB = (5 - -1) / (6 - 2)$

Must be y on top and subtraction (6 –2)

M1

$= \frac{6}{4} = \frac{3}{2} = 1.5$

Any correct equivalent

A1

2

(ii) $y - 5 = m(x - 6)$ or $y + 1 = m(x - 2)$

or $y = mx + c$ and attempt to find c

M1

$2y - 10 = 3x - 18$ etc leading to

$3x - 2y = 8$

AG *(be convinced)*

A1

2

(d) Attempt to eliminate x or y

$7x = 28$ etc

M1

$x = 4$

A1

$y = 2$

C is point (4,2)

A1

3

[12]

Q5.

(a) $mx - 1 = x^2 - 5x + 3$

Strict mark here – no trailing equals signs

$\Rightarrow x^2 - 5x - mx + 4 = 0$

$\Rightarrow x^2 - (5 + m)x + 4 = 0$

AG

(be convinved about algebra and = 0)

B1

1

(b) $(5 + m)^2 - 16 = 0$

Equal roots when "$b^2 - 4ac$" = 0 used

M1

$5 + m = (\pm)4$ or $(m + 1)(m + 9) = 0$

Square root or factor/formula attempt

m1

$m = -1$

A1

$m = -9$

***SC** B1, B1 only for –1, –9 (no working)*

A1

4

(c) Line is a tangent to the curve

Line touches curve, cuts at one point etc

E1

1

[6]